AP PRECALCULUS

Unit 1 • Lesson 1.5 • Notes (filled in)

Topic 1.5 • Polynomial Functions and Complex Zeros

Notes (filled in)

Learning goals

- Connect real zeros, linear factors and x-intercepts.
- Use multiplicity to predict the graph near each zero.
- Apply the Fundamental Theorem of Algebra and the conjugate pairs rule; write polynomials from their zeros.
- Determine whether a function is even, odd, or neither.

Vocabulary. zero, factor, multiplicity, complex number, imaginary unit i, conjugate, even function, odd function.

1. Zeros, factors and intercepts

Key idea

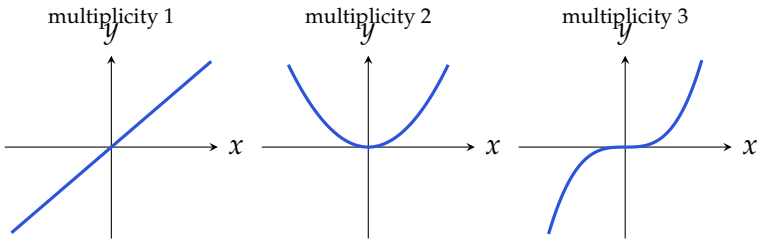
For a polynomial p and a real number a, these statements are equivalent: a is a __zero__ of p; $(x-a)$ is a __factor__ of $p(x)$; $(a,0)$ is an __x-intercept__ of the graph.

2. Multiplicity

If $(x-a)^k$ is a factor of $p(x)$ but $(x-a)^{k+1}$ is not, a is a zero of multiplicity __k__.

Key idea

Even multiplicity: the graph __touches the x-axis and turns back__.
Odd multiplicity: the graph __crosses the x-axis__; for multiplicity 3 or more it also __flattens__ as it crosses.



3. Complex zeros

Key idea

Fundamental Theorem of Algebra: a polynomial of degree n has exactly __n__ complex zeros, counted with multiplicity.
Conjugate pairs: if a polynomial with real coefficients has the zero $a+bi$, it also has the zero __$a-bi$__. Non-real zeros are __not__ x-intercepts.

Example 1 • Finding all zeros

Find all zeros of $p(x) = x^4 - 5x^2 - 36$.

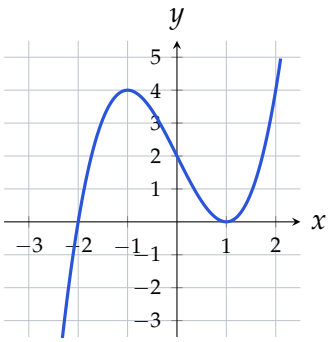
$x^4 - 5x^2 - 36 = (x^2 - 9)(x^2 + 4) = (x - 3)(x + 3)(x^2 + 4)$. Zeros: 3, -3 (real, x-intercepts) and $\pm 2i$ (non-real).

Example 2 • Building a polynomial

Write a polynomial of least degree with real coefficients, leading coefficient 1, and zeros -1 (multiplicity 2) and $2i$. Give its degree.

$-2i$ is also a zero. $p(x) = (x + 1)^2(x - 2i)(x + 2i) = (x + 1)^2(x^2 + 4)$, degree 4.

Example 3 • From a graph



The graph of a cubic q is shown. Write $q(x)$.

Crosses at -2 (odd multiplicity), touches at 1 (even): $q(x) = a(x + 2)(x - 1)^2$. The y-intercept is 2: $a \cdot 2 \cdot 1 = 2$, $a = 1$. So $q(x) = (x + 2)(x - 1)^2$.

4. Even and odd functions

Key idea

f is **even** if $f(-x) =$ $f(x)$ for every x; its graph is symmetric about the y-axis.
f is **odd** if $f(-x) =$ $-f(x)$; its graph is symmetric about the origin.

Example 4

Classify $f(x) = x^4 - 3x^2$, $g(x) = x^3 - 5x$, $h(x) = x^3 + 1$.

$f(-x) = x^4 - 3x^2 = f(x)$: even. $g(-x) = -x^3 + 5x = -g(x)$: odd. $h(-x) = -x^3 + 1$, equal to neither $h(x)$ nor $-h(x)$: neither.

Common mistakes

- Listing non-real zeros as x-intercepts.
- Writing the factor for the zero -3 as $(x - 3)$.
- Forgetting the conjugate when building a polynomial with real coefficients.

Check your understanding

1. A degree-4 polynomial with real coefficients can have 0, 2 or 4 real zeros (with multiplicity).
2. $f(x) = x^5 - 2x^3 + x$ is odd.